Hadamard Exact Model Protocol

This note answers the practical question:

`Can this be turned into a scan, or does it have to be tailored step by step?`

The answer is:

- yes, a large part of it can be turned into a repeatable scan
- no, the whole process cannot be reduced to one blind scan yet
- the right shape is a protocol with fixed stages plus adaptive branching

The Fixed Part

These steps are stable and should be treated as the default protocol:

1. Start from the best preserved basin checkpoint.
2. Run full GS/SDS defect reporting.
3. Read:
- total score
- max shift violation
- top unique SDS defects
- indicator Fourier deviation
4. Choose a monitored shift set.
5. Export a bounded PB/ILP instance.
6. Run the exact tier schedule.
7. Materialize any solved occupancy state as a real JSON artifact.
8. Run the full defect report on that real state.
9. If a local or exact run contains a better `best_score_seen`, promote that state into its own checkpoint.
10. Repeat from the promoted best real basin.

Those ten steps are the stable scan backbone.

The Adaptive Part

These decisions still need tailoring:

1. How wide should the monitored shift ring be?
2. Is the current surrogate honest enough?

- weighted absolute defects
- square-sum over monitored defects
- another cap

3. Did the exact solve clean the monitored ring but leak elsewhere?
4. Should the next rung be:

- bigger PB/ILP ring
- stronger surrogate
- bounded exact local repair
- or a mirror/oracle question?

So the process is not:

```
one export -&gt; one solve -&gt; done
```

It is:

```
export -&gt; solve -&gt; inspect leakage -&gt; adapt -&gt; repeat
```

What Should Be Automated

The following can be fully scripted:

- defect reporting
- shift ranking
- monitored ring export
- PB/ILP generation
- tier schedule generation
- solved-state materialization
- best-score-seen extraction into a standalone checkpoint
- comparison table against the prior basin

That means we can build a repeatable scan harness.

What Still Needs Judgment

The following still need operator or mirror judgment:

- whether the monitored ring is too narrow
- whether the exact model is “lying” by hiding leakage
- whether the surrogate is helping or distorting

- whether a better exact reduction is needed instead of another local cycle

This is the part that should be logged, not improvised from memory.

Protocol Diagram

```
flowchart TD
    A["Current best basin"] --> B["Full defect report"]
    B --> C["Choose monitored ring"]
    C --> D["Export PB/ILP"]
    D --> E["Run exact tier schedule"]
    E --> F["Materialize solved state"]
    F --> G["Full defect report on solved state"]
    G --> H{"Full score improved?"}
    H -- "Yes" --> I["Promote new basin"]
    H -- "No" --> J{"Did monitored ring improve?"}
    J -- "No" --> K["Change surrogate or move budget"]
    J -- "Yes" --> L["Add leakage ring"]
    K --> C
    L --> C
    I --> M{"Local plateau?"}
    M -- "Yes" --> N["Bounded exact local repair"]
    N --> O["Promote best_score_seen if better"]
    O --> B
    M -- "No" --> B
```

Mirror Use

The mirror should not be in every loop.

Use the mirror when:

- the exact model keeps improving the monitored ring but harming the full basin
- the same leakage pattern repeats
- a better constraint family is needed
- a new surrogate or reduction is needed

Do not use the mirror just to repeat the current rung verbatim.

The mirror is for:

- new reduction layers
- new invariants
- new exact surfaces

The local code should handle:

- repeated export
- repeated solve

- repeated materialization
- repeated compare

Decision Rule

Use this decision rule after every rung:

1. If full score improves, promote the new basin.
2. If full score does not improve but the monitored ring improves, widen the ring or strengthen the surrogate.
3. If exact-model output creates a promising intermediate state, hand it back into bounded local repair.
4. If the same failure repeats twice, ask the mirror for a new reduction instead of another blind rerun.

Current Best Practice

Right now the strongest protocol is:

1. report the full `668` basin
2. export bounded PB/ILP over a monitored ring
3. if monitored-only exact solve leaks, add a second ring
4. prefer a monitored square-sum surrogate over simple weighted absolute defects
5. if the exact-model state is cleaner but still not a new basin, hand it into bounded exact local repair
6. promote any `best_score_seen` basin that beats the current best endpoint

That is the cycle that produced the current local `2880` rung.